

The 4 Questions That Break Math Ph.D.s In Quant Interviews

Why textbook derivatives theory fails on the trading desk
— and what top market-making & vol desks actually look for.

01 // DELTA HEDGING

The "Risk-Free" Illusion

Gamma slippage & pin risk

02 // FLAT VOLATILITY

The Black-Scholes Trap

Trading Vanna & Volga

03 // PNL ATTRIBUTION

Straddle Realities

Realized vol vs Implied theta

04 // LOCAL VOLATILITY

The Dupire Breakdown

Why SLV models rule

SWIPE TO UNPACK >>

You are delta-hedged. The stock moves 4%. What is your exact PnL?

TEXTBOOK ANSWER

PnL is approximately zero because delta is zero.

Why it fails: The interviewer instantly marks you as an academic who has never observed a real trading book.

DESK REALITY

A delta-hedged book is NOT risk-free.

It is a pure bet on Realized Volatility vs. Implied Volatility.

$$d(\text{PnL}) \approx \frac{1}{2} \cdot \Gamma \cdot (dS)^2 + \Theta \cdot dt + \text{Cost}(\text{Rebalancing})$$

- In discrete time, you rebalance at intervals, suffering Gamma slippage.
- If the move is a discontinuous jump, your hedge fails to cover the gap.
- Near expiry, ATM options have exploding Gamma (pin risk).

PRACTITIONER 3-SECOND ANSWER:

Delta eliminates first-order directional drift. My PnL is determined by Gamma convexity gains versus Theta decay over the rebalancing window.

Why do market makers trade Vanna & Volga instead of single-option flat Vega?

TEXTBOOK ANSWER

Vega tells you price sensitivity to volatility σ . You just hedge total Vega.

The Trap: Assumes the volatility surface shifts uniformly across all strikes.

DESK REALITY

The Volatility Surface is not flat, and it never shifts in parallel.

Vanna = $\partial\Delta / \partial\sigma = \partial\text{Vega} / \partial S$ (Spot-Vol Cross Risk)

Tells you how your delta hedge changes when implied vol spikes or collapses.

Volga = $\partial\text{Vega} / \partial\sigma = \partial^2V / \partial\sigma^2$ (Vol Convexity)

Measures the curvature of options price wrt volatility. Essential for wing pricing.

WHY INTERVIEWERS LOVE THIS:

If you are long wings and short ATM, your net Vega might be zero, but you are massively long Volga and exposed to market skew shocks.

Walk me through the daily PnL of a long ATM Straddle held for 1 month.

TEXTBOOK ANSWER

A straddle profits if the stock makes a big move in either direction.

The Trap: True at expiration, but completely wrong for daily desk risk management.

DESK REALITY

A straddle is an ongoing race: Gamma profits vs Theta bleed.

$$\text{Daily PnL} \approx \frac{1}{2} \cdot S^2 \cdot \Gamma \cdot (\sigma_{\text{realized}}^2 - \sigma_{\text{implied}}^2) \cdot \Delta t$$

Black-Scholes identity: $\Theta = -\frac{1}{2} \cdot S^2 \cdot \Gamma \cdot \sigma_{\text{implied}}^2$

The 2 Trading Desk Rules:

1. If Realized Vol > Implied Vol: Gamma profits exceed Theta decay. You win.
2. If Realized Vol < Implied Vol: The stock moves, but not enough to offset Theta.

Even if the stock rises 5%, you bleed money if realized vol was below implied.

CANDIDATE DIFFERENTIATOR:

Speak in terms of volatility spread (Realized σ vs Implied σ),
not naive directional price targets.

Why can't you use Dupire Local Vol to price path-dependent options (e.g. Cliquets)?

TEXTBOOK ANSWER

Dupire fits European vanilla prices exactly, so it must price exotics correctly.

The Trap: Fitting static marginal distributions does NOT determine path dynamics.

DESK REALITY

Dupire flattens future forward smiles.

Under a deterministic local vol diffusion $dS_t = r S_t dt + \sigma(t, S_t) S_t dW_t$:

The Forward Volatility Flaw:

As time rolls forward, the conditional distribution of spot flattens out, severely underpricing forward skew, forward-starting options, & cliquets.

The Production Fix on Desks:

Stochastic Local Volatility (SLV) — combining Heston stochastic vol with a non-parametric leverage function to fit both smile & forward skew.

WHAT INTERVIEWERS ARE TESTING:

Do you understand model risk? An arbitrage-free static fit does NOT protect you when hedging path-dependent exotic payouts.

Stop Studying Like a Student. Prepare Like a Trading Desk.

ABSOLUTE BEGINNERS

Quant Finance for Absolute Beginners

- 378 pages across 28 structured chapters from the ground up
- 28 runnable Python scripts covering Itô, pricing, and Greeks (Code: BEGINNER10)

INTERVIEW CANDIDATES

Common Mistakes in Quant Interviews

- 106 pages covering 29 failure modes that kill interview performance
- Wrong Answer → Trap → Correction → 3-Second Answer (Code: MISTAKES10)

DERIVATIVES DESKS

Vol Surface Construction & Derivatives Pack

- SSVI smile calibration, Dupire local vol surfaces, and 0DTE options dynamics
- Discrete delta-hedging Jupyter simulation notebooks included

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